

Roll No.

TAS-801

**B. TECH. (ME) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023**

AIRPORT AND AIRLINE MANAGEMENT

(Elective)

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Explain the unique features of *Wright Flyer*, the first successful aircraft developed by the Wright Brothers in 1903. (CO-K2)
- (b) Differentiate between ICAO and IATA based on their mandate and functioning. (CO-K2)
- (c) Describe the top five challenges being faced by the global aviation industry in the post-Covid-19 era. (CO-K2)
2. (a) Describe the various components of Airside and Landside area of a typical Commercial airport. (CO-K3)

P. T. O.

(2)

- (b) Describe the organizational structure of a typical International Airport dealing with Passengers and Cargo. (CO-K3)
- (c) What is the difference between air express, air freight, and air mail service ? (CO-K3)
- 3. (a) Elaborate upon the merits and demerits of PPP for development of new Airports in India. (CO-K4)
- (b) List down the breakup and percentage of revenue earned by a typical Airport that deals with International Airlines and Cargo. (CO-K4)
- (c) What are the key environmental regulations that are applicable for development of a Greenfield Airport in India ? (CO-K4)
- 4. (a) What are the key precautions to be taken while handling unaccompanied minors and disabled passengers at an Airport ? (CO-K4)
- (b) What are the precautions to be taken while handling passengers on Stretchers and Human Remains ? (CO-K4)
- (c) Differentiate between Customs and Immigrations formalities and procedures based on their applicability and purpose. (CO-K4)
- 5. (a) Elaborate upon the role of a warehouse in ensuring proper supply chain management. Highlight the key challenges in operating a warehouse at an International Airport. (CO-K4)
- (b) What are the various types of Cargo that are handled by an International Airline ? Discuss the challenges in handling each type of Cargo. (CO-K4)
- (c) Explain the term Turn Around Time (TAT) of an Aircraft and describe some measures by which it can be reduced for a low-cost Airline. (CO-K4)

Roll No.

TME-812

B. TECH. (ME) (EIGHTH SEMESTER)

END SEMESTER EXAMINATION, May, 2023

NON-CONVENTIONAL ENERGY RESOURCES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain energy exploitation in detail. (CO1)
(b) Explain different non-conventional energy resources. (CO1)
(c) Explain energy demand analysis in detail. (CO1)
2. (a) What is solar radiation? Explain the working principle of flat plate solar collector with neat sketch. (CO2)

P. T. O.

- (b) Derive the expression for temperature at the outlet of compound parabolic concentrator (CPC) and hence derive the expression for heat gain for CPC. (CO2)
- (c) Explain different sensible heat storage methods with neat sketch. (CO2)
3. (a) What is wind energy? Write the property of wind energy. Also, explain different parameters for selection of site for a wind mill. (CO3)
- (b) A propelling type of wind turbine has the following data : (CO3)
- Speed of wind at the height of 10 m = 12 m/s, Air density = 1.226 kg/m^3 , Alpha = 0.14, Height of tower = 100 m, Diameter of rotor = 80 m, Wind speed at the turbine reduces by 20% and generator efficiency = 85%. Find (i) total power available in wing, (ii) power extracted by the turbine, (iii) electrical power generated, (iv) axial thrust on the turbine and (v) maximum axial thrust on the turbine.
- Assume suitable data, if required. (CO3)
- (c) Explain different types of geothermal station with schematic representation. (CO3)
4. (a) What do you mean by tidal power? Explain the different components of tidal power plant. Write the limitations of tidal energy conversion systems. (CO4, CO5)
- (b) A tidal power plant of single basin type has basin area of 24 km^2 . The tide has a range of 10 km. The turbine stops operation when the head on it falls below 3. Calculate the average power generated during one

(3)

filling/emptying in MW if the turbine generator efficiency is 75%.

Density of sea water = 1025 kg/m^3 .

Assume suitable data, if required.

(CO4, CO5)

- (c) What are the different methods of production of hydrogen ? Explain in short with neat sketch. (CO4, CO5)

5. (a) What do you mean by energy audit ? Explain the energy audit for boilers and turbines. (CO3, CO6)

- (b) Write in detail the various applications of hydrogen energy. (CO3, CO6)

- (c) Explain the different types of horizontal axis wind turbine power plant.

(CO3, CO6)

Roll No.

TME-814

B. TECH. (ME) (EIGHTH SEMESTER)
END SEMESTER EXAMINATION, May, 2023
ADVANCED WELDING TECHNOLOGY

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) Compare TIG and MIG welding processes. (CO1)
(b) Describe the principle of Resistance welding with neat sketch. (CO1)
(c) Classify Electrodes. What are the functions of electrode coatings ? (CO1)
2. (a) Describe Explosive Welding process and its benefits. (CO2)
(b) Describe Electron Beam welding process and its limitation with sketch. (CO2)
(c) Describe under water welding process and its benefits. (CO2)
3. (a) What is carbon equivalent ? Indicate its application. (CO3)

P. T. O.

(2)

- (b) Describe the types of distortion in welding. What are the steps taken to control the same ? (CO3)
- (c) Explain hot and cold cracking. How can they be controlled ? (CO3)
4. (a) Explain the following : (CO4)
- (i) Cathode spot
 - (ii) Anode spot
 - (iii) Arc column
 - (iv) Ionization potential
- (b) What are different welding defects ? Explain their cause of occurrence. (CO4)
- (c) Explain the formation of Heat Affected Zone in the welding joints. (CO4)
5. (a) How is X-ray testing done of welded joints ? Explain with sketch. (CO4, CO5, CO6)
- (b) Explain Ultrasonic and Magnetic particle tests conducted for welding. (CO4, CO5, CO6)
- (c) Explain in brief the physics of welding arc. (CO4, CO5, CO6)